Hartree-Fock-Nambu mean-field theory

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A. Introduction

Consider an interacting many-body system of electrons described by the Hamiltonian

$$H = H_0 + H_{\text{int}} = \sum_{\alpha\beta} h_{\alpha\beta} c^\dagger_\alpha c_\beta + \sum_{\alpha\beta\gamma\delta} V^{\alpha\beta}_{\gamma\delta} c^\dagger_\alpha c^\dagger_\beta c_\gamma c_\delta, \tag{1}$$

with $c^\dagger(c)$ fermionic creation (annihilation) operators and the Greek indices α, β, γ, δ generic degrees of freedom. The time-dependency on the fermionic operators are being omitted for clarity, as is also done in the non-interacting Hamiltonian, accounting for a coupling to a perturbative external force, i.e $h_{\alpha\beta} \to h_{\alpha\beta} + B_{\alpha\beta}F(t)$. Concerning the interaction term, we assume $V(\mathbf{r})$ to be generic electrostatic potential of symmetry $V(\mathbf{r}-\mathbf{r}') = V(\mathbf{r}'-\mathbf{r})$ such that the interacting potential tensor has the symmetries $V^{\alpha\beta}_{\gamma\delta} = V^{\beta\alpha}_{\delta\gamma}$ and $(V^{\alpha\beta}_{\gamma\delta})^* = V^{\gamma\delta}_{\alpha\beta}$.

Symmetry properties of V
From the imposed symmetry $V(\mathbf{r}-\mathbf{r}') = V(\mathbf{r}'-\mathbf{r})$, we obtain

$$\begin{aligned} V^{\alpha\beta}_{\gamma\delta} &= \int d\mathbf{r}d\mathbf{r}' \phi^*_\alpha(\mathbf{r})\phi^*_\beta(\mathbf{r}')V(\mathbf{r}-\mathbf{r}')\phi_\gamma(\mathbf{r}')\phi_\delta(\mathbf{r}),\ \mathbf{r} \leftrightharpoons \mathbf{r}' \\ &= \int d\mathbf{r}d\mathbf{r}' \phi^*_\alpha(\mathbf{r}')\phi^*_\beta(\mathbf{r})V(\mathbf{r}-\mathbf{r}')\phi_\gamma(\mathbf{r})\phi_\delta(\mathbf{r}') \\ &= \int d\mathbf{r}d\mathbf{r}' \phi^*_\beta(\mathbf{r})\phi^*_\alpha(\mathbf{r}')\, V(\mathbf{r}-\mathbf{r}')\phi_\delta(\mathbf{r}')\phi_\gamma(\mathbf{r}) = V^{\beta\alpha}_{\delta\gamma} \ \square \end{aligned}$$

Similarly, we can show that

$$\begin{aligned} \left(V^{\alpha\beta}_{\gamma\delta}\right)^* &= \int d\mathbf{r}d\mathbf{r}' \phi_\alpha(\mathbf{r})\phi_\beta(\mathbf{r}')V(\mathbf{r}-\mathbf{r}')\phi^*_\gamma(\mathbf{r}')\phi^*_\delta(\mathbf{r}) \\ &= \int d\mathbf{r}d\mathbf{r}' \phi^*_\delta(\mathbf{r})\phi^*_\gamma(\mathbf{r}')V(\mathbf{r}-\mathbf{r}')\phi_\delta(\mathbf{r}')\phi_\alpha(\mathbf{r}) = V^{\delta\gamma}_{\beta\alpha} = V^{\gamma\delta}_{\alpha\beta} \ \square \end{aligned}$$

In order to study the dynamics of this system, we solve for the time-evolution of the the system's reduced density matrix (rDM) through it's its equation of motion (EoM). The system's purely electronic reduced density matrix (rDM) is defined as $\rho^{\text{ee}}_{ba} = \langle c^\dagger_a c_b \rangle$. However, if one wishes to study superconducting systems, one must also account for the anomalous terms. For this, we introduce the Nambu spinor representation with doubling of degrees of freedom $\check{c}_a = (c_a\ c^\dagger_a)$. The Nambu representation of the rDM, can then be written as the tensor product of this spinors for each of the spin combinations,

$$\rho_{ab} = \begin{pmatrix} \langle c^\dagger_b c_a \rangle & \langle c_b c_a \rangle \\ \langle c^\dagger_b c^\dagger_a \rangle & \langle c_b c^\dagger_a \rangle \end{pmatrix} = \begin{pmatrix} \rho^{\text{ee}}_{ab} & \rho^{\text{eh}}_{ab} \\ \rho^{\text{he}}_{ab} & \rho^{\text{hh}}_{ab} \end{pmatrix}. \tag{2}$$

Directly from the (equal-time) fermionic anti-commutator properties, this terms relate to each other as $\rho^{\text{hh}}_{ab} = -\rho^{\text{ee}}_{ba} + \delta_{ab}$ and $\rho^{\text{he}}_{ab} = (\rho^{\text{eh}}_{ba})^*$, but also $\rho^{\text{ee/hh}}_{ab} = (\rho^{\text{ee/hh}}_{ba})^*$ and $\rho^{\text{eh/he}}_{ba} = -\rho^{\text{eh/he}}_{ab}$. Notice that, since the spin degree of freedom was not yet specified we are not yet restricting ourselves to typical s-wave spin-singlet superconductors, this is a general treatment. Our task now is then to solve the two distinct EoMs, for the purely electronic rDM ρ^{ee}_{ab}, which then determines the other two simple through their relations.

B. Reduced density matrices equations of motion

We start by solving for the purely electronic rDM equation of motions

$$\frac{d}{dt}\rho^{\text{ee}}_{ab}(t) = \left\langle \frac{dc^\dagger_b(t)}{dt} c_a(t) \right\rangle + \left\langle c^\dagger_b(t) \frac{dc_a(t)}{dt} \right\rangle, \tag{3}$$

in the Heisenberg picture of quantum mechanics, where the fermionic operators evolve accordingly to the Heisenberg equation. Therefore the rDM equation of motion reads

$$\frac{d}{dt}\rho^{\text{ee}}_{ab} = \left\langle \frac{i}{\hbar}[H, c^\dagger_b] c_a \right\rangle + \left\langle c^\dagger_b \frac{i}{\hbar}[H, c_a] \right\rangle. \tag{4}$$

where we have omitted the time dependency in the fermionic operators for compactness. This practice is done throughout the remaining work with no further mention. As shown in Appendix, given the equal-time fermionic anti-commutator properties $\left\{c_a, c_b^\dagger\right\} = \left\{c_b^\dagger, c_a\right\} = \delta_{ab}$ and $\{c_a, c_b\} = \left\{c_b^\dagger, c_a^\dagger\right\} = 0$, each of the commutators are evaluated to

$$[H, c_b^\dagger] = \bar{h}_{\alpha b} c_\alpha^\dagger + \frac{1}{4}\left(V_{\gamma b}^{\alpha\beta} - V_{b\gamma}^{\alpha\beta}\right)\left(c_\alpha^\dagger c_\beta^\dagger c_\gamma - c_\gamma c_\beta^\dagger c_\alpha^\dagger\right) \tag{5}$$

$$[H, c_a] = -\bar{h}_{a\alpha} c_\alpha + \frac{1}{4}\left(V_{\beta\gamma}^{\alpha a} - V_{\beta\gamma}^{a\alpha}\right)\left(c_\alpha^\dagger c_\beta c_\gamma - c_\gamma c_\beta c_\alpha^\dagger\right), \tag{6}$$

where we have hidden the summations on repeated indices using Einstein's notation.

Commutators calculations

First we calculate the commutator with $c_b^\dagger$ making use of the following commutator properties. We obtain

$$\left[c_\alpha^\dagger c_\beta, c_b^\dagger\right] = c_\alpha^\dagger \left\{c_\beta, c_b^\dagger\right\} - \cancel{\left\{c_b^\dagger, c_\alpha^\dagger\right\}} c_\beta = c_\alpha^\dagger \delta_{\beta b}$$
$$\left[c_\beta c_\alpha^\dagger, c_b^\dagger\right] = c_\beta \cancel{\left\{c_\alpha^\dagger, c_b^\dagger\right\}} - \left\{c_b^\dagger, c_\beta\right\} c_\alpha^\dagger = -c_\alpha^\dagger \delta_{\beta b}$$

and that

$$\left[c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta, c_b^\dagger\right] = c_\alpha^\dagger c_\beta^\dagger \left(c_\gamma \left\{c_\delta, c_b^\dagger\right\} - \left\{c_b^\dagger, c_\gamma\right\} c_\delta\right) + \left(c_\alpha^\dagger \cancel{\left\{c_\beta^\dagger, c_b^\dagger\right\}} - \cancel{\left\{c_b^\dagger, c_\alpha^\dagger\right\}} c_\beta^\dagger\right) c_\gamma c_\delta = c_\alpha^\dagger c_\beta^\dagger c_\gamma \delta_{\delta b} - c_\alpha^\dagger c_\beta^\dagger c_\delta \delta_{\gamma b}$$
$$\left[c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger, c_b^\dagger\right] = c_\delta c_\gamma \left(c_\beta^\dagger \cancel{\left\{c_\alpha^\dagger, c_b^\dagger\right\}} - \cancel{\left\{c_b^\dagger, c_\beta^\dagger\right\}} c_\alpha^\dagger\right) + \left(c_\delta \left\{c_\gamma, c_b^\dagger\right\} - \left\{c_b^\dagger, c_\delta\right\} c_\gamma\right) c_\beta^\dagger c_\alpha^\dagger = c_\delta c_\beta^\dagger c_\alpha^\dagger \delta_{\gamma b} - c_\gamma c_\beta^\dagger c_\alpha^\dagger \delta_{\delta b}$$

The complete commutator then reads

$$\begin{aligned}
[H, c_b^\dagger] &= \left[\frac{1}{2}\bar{h}_{\alpha\beta}\left(c_\alpha^\dagger c_\beta - c_\beta c_\alpha^\dagger\right) + \frac{1}{4}V_{\gamma\delta}^{\alpha\beta}\left(c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta + c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger\right) + \frac{1}{2}\cancel{\tilde{h}_0}, c_b^\dagger\right] \\
&= \frac{1}{2}\bar{h}_{\alpha\beta}\left(\left[c_\alpha^\dagger c_\beta, c_b^\dagger\right] - \left[c_\beta c_\alpha^\dagger, c_b^\dagger\right]\right) + \frac{1}{4}V_{\gamma\delta}^{\alpha\beta}\left(\left[c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta, c_b^\dagger\right] + \left[c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger, c_b^\dagger\right]\right) \\
&= \bar{h}_{\alpha\beta} c_\alpha^\dagger \delta_{\beta b} + \frac{1}{4}\left(V_{\gamma\delta}^{\alpha\beta} c_\alpha^\dagger c_\beta^\dagger c_\gamma \delta_{\delta b} - V_{\gamma\delta}^{\alpha\beta} c_\alpha^\dagger c_\beta^\dagger c_\delta \delta_{\gamma b} + V_{\gamma\delta}^{\alpha\beta} c_\delta c_\beta^\dagger c_\alpha^\dagger \delta_{\gamma b} - V_{\gamma\delta}^{\alpha\beta} c_\gamma c_\beta^\dagger c_\alpha^\dagger \delta_{\delta b}\right) \\
&= \bar{h}_{\alpha b} c_\alpha^\dagger + \frac{1}{4}\left(V_{\gamma b}^{\alpha\beta} - V_{b\gamma}^{\alpha\beta}\right)\left(c_\alpha^\dagger c_\beta^\dagger c_\gamma - c_\gamma c_\beta^\dagger c_\alpha^\dagger\right)
\end{aligned}$$

Analogous for the $[H, c_a]$ commutator, we have

$$\left[c_\alpha^\dagger c_\beta, c_a\right] = c_\alpha^\dagger \cancel{\{c_\beta, c_a\}} - \left\{c_a, c_\alpha^\dagger\right\} c_\beta = -c_\beta \delta_{a\alpha}$$
$$\left[c_\beta c_\alpha^\dagger, c_a\right] = c_\beta \left\{c_\alpha^\dagger, c_a\right\} - \cancel{\{c_a, c_\beta\}} c_\alpha^\dagger = c_\beta \delta_{a\alpha}$$

and that

$$\left[c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta, c_a\right] = c_\alpha^\dagger c_\beta^\dagger \left(c_\gamma \cancel{\{c_\delta, c_a\}} - \cancel{\{c_a, c_\gamma\}} c_\delta\right) + \left(c_\alpha^\dagger \left\{c_\beta^\dagger, c_a\right\} - \left\{c_a, c_\alpha^\dagger\right\} c_\beta^\dagger\right) c_\gamma c_\delta = c_\alpha^\dagger c_\gamma c_\delta \delta_{\beta a} - c_\beta^\dagger c_\gamma c_\delta \delta_{a\alpha}$$
$$\left[c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger, c_a\right] = c_\delta c_\gamma \left(c_\beta^\dagger \left\{c_\alpha^\dagger, c_a\right\} - \left\{c_a, c_\beta^\dagger\right\} c_\alpha^\dagger\right) + \left(c_\delta \cancel{\{c_\gamma, c_a\}} - \cancel{\{c_a, c_\delta\}} c_\gamma\right) c_\beta^\dagger c_\alpha^\dagger = c_\delta c_\gamma c_\beta^\dagger \delta_{\alpha a} - c_\delta c_\gamma c_\alpha^\dagger \delta_{a\beta}$$

The complete commutator then reads

$$\begin{aligned}
[H, c_a] &= \left[\frac{1}{2}\bar{h}_{\alpha\beta}\left(c_\alpha^\dagger c_\beta - c_\beta c_\alpha^\dagger\right) + \frac{1}{4}V_{\gamma\delta}^{\alpha\beta}\left(c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta + c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger\right) + \frac{1}{2}\cancel{\tilde{h}_0}, c_a\right] \\
&= \frac{1}{2}\bar{h}_{\alpha\beta}\left(\left[c_\alpha^\dagger c_\beta, c_a\right] - \left[c_\beta c_\alpha^\dagger, c_a\right]\right) + \frac{1}{4}V_{\gamma\delta}^{\alpha\beta}\left(\left[c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta, c_a\right] + \left[c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger, c_a\right]\right) \\
&= -\bar{h}_{\alpha\beta} c_\beta \delta_{a\alpha} + \frac{1}{4}\left(V_{\gamma\delta}^{\alpha\beta} c_\alpha^\dagger c_\gamma c_\delta \delta_{\beta a} - V_{\gamma\delta}^{\alpha\beta} c_\beta^\dagger c_\gamma c_\delta \delta_{a\alpha} + V_{\gamma\delta}^{\alpha\beta} c_\delta c_\gamma c_\beta^\dagger \delta_{\alpha a} - V_{\gamma\delta}^{\alpha\beta} c_\delta c_\gamma c_\alpha^\dagger \delta_{a\beta}\right) \\
&= -\bar{h}_{a\beta} c_\beta + \frac{1}{4}\left(V_{\gamma\delta}^{\alpha a} c_\alpha^\dagger c_\gamma c_\delta - V_{\gamma\delta}^{a\beta} c_\beta^\dagger c_\gamma c_\delta + V_{\gamma\delta}^{a\beta} c_\delta c_\gamma c_\beta^\dagger - V_{\gamma\delta}^{\alpha a} c_\delta c_\gamma c_\alpha^\dagger\right) \\
&= -\bar{h}_{a\alpha} c_\alpha + \frac{1}{4}\left(V_{\beta\gamma}^{\alpha a} - V_{\beta\gamma}^{a\alpha}\right)\left(c_\alpha^\dagger c_\beta c_\gamma - c_\gamma c_\beta c_\alpha^\dagger\right)
\end{aligned}$$

$$\{c_a, c_b^\dagger\} = \{c_b^\dagger, c_a\} = \delta_{ab}$$
$$\{c_a, c_b\} = \{c_b^\dagger, c_a^\dagger\} = 0$$

$$[AB, Z] = A\{B, Z\} - \{Z, A\}B$$
$$[AB, Z] = A[B, Z] - [Z, A]B$$
$$[AB, CD] = -A(C\{D, B\} - \{B, C\}D) - (C\{D, A\} - \{A, C\}D)B$$
$$[ABCD, Z] = AB(C\{D, Z\} - \{Z, C\}D) + (A\{B, Z\} - \{Z, A\}B)CD$$

Substituting Eqs.(5) and (6) back into Eq.(4) yields

$$\begin{aligned} -i\hbar\frac{d}{dt}\rho_{ab}^{\mathrm{ee}} &= \rho_{a\alpha}^{\mathrm{ee}}\bar{h}_{\alpha b} - \bar{h}_{a\alpha}\rho_{\alpha b}^{\mathrm{ee}} \\ &+ \frac{1}{4}\left(V_{\gamma b}^{\alpha\beta} - V_{b\gamma}^{\alpha\beta}\right)\left(\left\langle c_\alpha^\dagger c_\beta^\dagger c_\gamma c_a\right\rangle - \left\langle c_\gamma c_\beta^\dagger c_\alpha^\dagger c_a\right\rangle\right) \\ &+ \frac{1}{4}\left(V_{\beta\gamma}^{\alpha a} - V_{\beta\gamma}^{a\alpha}\right)\left(\left\langle c_b^\dagger c_\alpha^\dagger c_\beta c_\gamma\right\rangle - \left\langle c_b^\dagger c_\gamma c_\beta c_\alpha^\dagger\right\rangle\right) \end{aligned} \tag{7}$$

where we came back to the definition of the normal rDM in Eq.(**??**) for the first two terms. Notice, however, that the remaining interaction terms correspond to the expectation values of four-operators. For this, we introduce a mean-field approximation where we assume the two-particle expectation value to simply behave as a product of two one-particle expectation values. From this mean-field decoupling we can then make use of *Wick's theorem* as done in the Appendix. We obtain

$$\left\langle c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta\right\rangle \approx \rho_{\beta\alpha}^{\mathrm{he}}\rho_{\delta\gamma}^{\mathrm{eh}} - \rho_{\gamma\alpha}^{\mathrm{ee}}\rho_{\delta\beta}^{\mathrm{hh}} + \rho_{\delta\alpha}^{\mathrm{ee}}\rho_{\gamma\beta}^{\mathrm{ee}} \tag{8}$$

$$\left\langle c_\alpha c_\beta^\dagger c_\gamma^\dagger c_\delta\right\rangle \approx \rho_{\beta\alpha}^{\mathrm{hh}}\rho_{\delta\gamma}^{\mathrm{ee}} - \rho_{\gamma\alpha}^{\mathrm{hh}}\rho_{\delta\beta}^{\mathrm{ee}} + \rho_{\delta\alpha}^{\mathrm{eh}}\rho_{\gamma\beta}^{\mathrm{he}} \tag{9}$$

$$\left\langle c_\alpha^\dagger c_\beta c_\gamma c_\delta^\dagger\right\rangle \approx \rho_{\beta\alpha}^{\mathrm{ee}}\rho_{\delta\gamma}^{\mathrm{ee}} - \rho_{\gamma\alpha}^{\mathrm{ee}}\rho_{\delta\beta}^{\mathrm{hh}} + \rho_{\delta\alpha}^{\mathrm{he}}\rho_{\gamma\beta}^{\mathrm{eh}} \tag{10}$$

As derived below in the gray appendices, the purely electronic rDM equation of motion can then be written in a time-dependent Hartree-Fock in a more compact and elegant manner as

$$i\hbar\frac{d}{dt}\rho_{ab}^{\mathrm{ee}} = \left[h_{ab}^{\mathrm{HF}}, \rho_{ab}^{\mathrm{ee}}\right] - \rho_{ab}^{\mathrm{ee}}\Delta_{ba}^* + \Delta_{ab}\rho_{ab}^{\mathrm{he}} \tag{11}$$

with the Hartree-Fock Hamiltonian being

$$h_{ab}^{HF}(\rho) = h_{ab} + \Sigma_{ab}^H(\rho) + \Sigma_{ab}^F(\rho), \tag{12}$$

identifying the Hartree, Fock and pairing self-energies (themselves dependent on the rDM) respectively as

$$\begin{aligned} \Sigma_{ab}^H &= \textstyle\sum_{\gamma\delta} V_{b\gamma}^{\delta a}\rho_{\gamma\delta}^{\mathrm{ee}} \\ \Sigma_{ab}^F &= -\textstyle\sum_{\gamma\delta} V_{b\gamma}^{a\delta}\rho_{\gamma\delta}^{\mathrm{ee}} \end{aligned} \quad \text{and} \quad \Delta_{ab} = \sum_{\gamma\delta} V_{\delta\gamma}^{ab}\rho_{\gamma\delta}^{\mathrm{eh}} \tag{13}$$

As a side note concerning excitons, notice that the Fock self-energy term in is an attractive interaction (due the negative sign) such that it lower the total energy of the system enabling the formation of the electron-hole bound states. Conversely, the Hartree self-energy term will control details of the excitation spectrum, such as spin-splitting. Also, note the symmetry $\Delta_{ab} = -\Delta_{ba}$.

Analogously for the anomalous EoM we have that

$$\frac{d}{dt}\rho_{ab}^{\mathrm{eh}} = \left\langle \frac{i}{\hbar}[H, c_b]c_a\right\rangle - \left\langle c_b\frac{i}{\hbar}[H, c_a]\right\rangle. \tag{14}$$

but instead perform the mean-field decoupling

$$\left\langle c_\alpha^\dagger c_\beta c_\gamma c_\delta\right\rangle \approx \rho_{\beta\alpha}^{\mathrm{ee}}\rho_{\delta\gamma}^{\mathrm{eh}} - \rho_{\gamma\alpha}^{\mathrm{eh}}\rho_{\delta\beta}^{\mathrm{eh}} + \rho_{\delta\alpha}^{\mathrm{ee}}\rho_{\gamma\beta}^{\mathrm{eh}}, \tag{15}$$

$$\left\langle c_\alpha c_\beta^\dagger c_\gamma c_\delta\right\rangle \approx \rho_{\beta\alpha}^{\mathrm{he}}\rho_{\delta\gamma}^{\mathrm{eh}} - \rho_{\gamma\alpha}^{\mathrm{eh}}\rho_{\delta\beta}^{\mathrm{ee}} + \rho_{\delta\alpha}^{\mathrm{eh}}\rho_{\gamma\beta}^{\mathrm{ee}}. \tag{16}$$

$$\left\langle c_\alpha c_\beta c_\gamma^\dagger c_\delta\right\rangle \approx \rho_{\beta\alpha}^{\mathrm{eh}}\rho_{\delta\gamma}^{\mathrm{ee}} - \rho_{\gamma\alpha}^{\mathrm{hh}}\rho_{\delta\beta}^{\mathrm{eh}} + \rho_{\delta\alpha}^{\mathrm{eh}}\rho_{\gamma\beta}^{\mathrm{hh}} \tag{17}$$

$$\left\langle c_\alpha c_\beta c_\gamma c_\delta^\dagger\right\rangle \approx \rho_{\beta\alpha}^{\mathrm{eh}}\rho_{\delta\gamma}^{\mathrm{hh}} - \rho_{\gamma\alpha}^{\mathrm{eh}}\rho_{\delta\beta}^{\mathrm{hh}} + \rho_{\delta\alpha}^{\mathrm{hh}}\rho_{\gamma\beta}^{\mathrm{eh}} \tag{18}$$

which yields the anomalous equation of motion

$$i\hbar\frac{d}{dt}\rho^{\mathrm{eh}}_{ab} = h^{\mathrm{HF}}_{ab}\,\rho^{\mathrm{eh}}_{ab} - \rho^{\mathrm{eh}}_{ab}h^{\mathrm{HF}}_{ba} + \Delta_{ab}\rho^{\mathrm{hh}}_{ab} - \rho^{\mathrm{ee}}_{ab}\Delta_{ab} \tag{19}$$

rDM EoM calculations

$$[H, c^\dagger_b] = h_{\alpha b}c^\dagger_\alpha + \frac{1}{2}(V^{\alpha\beta}_{\gamma b} - V^{\alpha\beta}_{b\gamma})c^\dagger_\alpha c^\dagger_\beta c_\gamma$$
$$[H, c_a] = -h_{a\alpha}c_\alpha + \frac{1}{2}(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma})c^\dagger_\alpha c_\beta c_\gamma$$

Using the mean-field decoupling in Eq.(**??**), the *normal* EoM interaction terms becomes

$$-\frac{1}{2}(V^{\alpha\beta}_{\gamma b} - V^{\alpha\beta}_{b\gamma})(\rho^{\mathrm{he}}_{\beta\alpha}\rho^{\mathrm{eh}}_{a\gamma} - \rho^{\mathrm{ee}}_{\gamma\alpha}\rho^{\mathrm{ee}}_{a\beta} + \rho^{\mathrm{ee}}_{a\alpha}\rho^{\mathrm{ee}}_{\gamma\beta})$$
$$-\frac{1}{2}(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma})(\rho^{\mathrm{he}}_{\alpha b}\rho^{\mathrm{eh}}_{\gamma\beta} - \rho^{\mathrm{ee}}_{\beta b}\rho^{\mathrm{ee}}_{\gamma\alpha} + \rho^{\mathrm{ee}}_{\gamma b}\rho^{\mathrm{ee}}_{\beta\alpha})$$

with the purely electronic reading

$$-\rho^{\mathrm{ee}}_{a\beta}(V^{\alpha\beta}_{b\gamma} - V^{\beta\alpha}_{b\gamma})\rho^{\mathrm{ee}}_{\gamma\alpha} + (V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma})\rho^{\mathrm{ee}}_{\gamma\alpha}\rho^{\mathrm{ee}}_{\beta b}$$
$$= -\rho^{\mathrm{ee}}_{a\beta}\Sigma^{\mathrm{HF}}_{\beta b} + \Sigma^{\mathrm{HF}}_{a\beta}\,\rho^{\mathrm{ee}}_{\beta b} = [\Sigma_{\mathrm{HF}}, \rho_{\mathrm{ee}}]|_{ab}$$

(no index is a matrix since I reserve the bold specifically to denote the spin structure) and the anomalous terms reads

$$-\frac{1}{2}\rho^{\mathrm{eh}}_{a\gamma}(V^{\beta\alpha}_{b\gamma} - V^{\beta\alpha}_{\gamma b})(\rho^{\mathrm{eh}}_{\alpha\beta})^* - \frac{1}{2}(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma})\rho^{\mathrm{eh}}_{\gamma\beta}\rho^{\mathrm{he}}_{\alpha b}$$
$$= -\frac{1}{2}\rho^{\mathrm{eh}}_{a\gamma}(\Delta_{b\gamma} - \Delta_{\gamma b})^* - \frac{1}{2}(\Delta_{\alpha a} - \Delta_{a\alpha})\rho^{\mathrm{he}}_{\alpha b}$$
$$= \rho_{\mathrm{eh}}\cdot\Delta^*|_{ab} + \Delta\cdot\rho_{\mathrm{he}}|_{ab}$$

The non-interacting terms is trivially $-[h, \rho_{\mathrm{ee}}]_{ab}$. Analogously for the *anomalous* EoM, we have trivially non-interacting term $-[\rho_{\mathrm{eh}}\cdot h^T]_{ab} - [h\cdot\rho_{\mathrm{eh}}]_{ab}$. For the interaction terms we perform instead the mean-field decoupling of $\langle c^\dagger_\alpha c_\beta c_\gamma c_a\rangle$ and $\langle c_b c^\dagger_\alpha c_\beta c_\gamma\rangle$, we obtain

$$-\frac{1}{2}(V^{\alpha b}_{\beta\gamma} - V^{b\alpha}_{\beta\gamma})(\rho^{\mathrm{ee}}_{\beta\alpha}\rho^{\mathrm{eh}}_{a\gamma} - \rho^{\mathrm{ee}}_{\gamma\alpha}\rho^{\mathrm{eh}}_{a\beta} + \rho^{\mathrm{ee}}_{a\alpha}\rho^{\mathrm{eh}}_{\gamma\beta})$$
$$-\frac{1}{2}(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma})(\rho^{\mathrm{hh}}_{\alpha b}\rho^{\mathrm{eh}}_{\gamma\beta} - \rho^{\mathrm{eh}}_{\beta b}\rho^{\mathrm{ee}}_{\gamma\alpha} + \rho^{\mathrm{eh}}_{\gamma b}\rho^{\mathrm{ee}}_{\beta\alpha})$$

The first term reading

$$\rho^{\mathrm{eh}}_{a\gamma}(V^{\alpha b}_{\gamma\beta} - V^{b\alpha}_{\gamma\beta})\rho^{\mathrm{ee}}_{\beta\alpha} - \frac{1}{2}\rho^{\mathrm{ee}}_{a\alpha}(V^{\alpha b}_{\beta\gamma} - V^{b\alpha}_{\beta\gamma})\rho^{\mathrm{eh}}_{\gamma\beta}$$
$$=\rho^{\mathrm{eh}}_{a\gamma}\Sigma^{\mathrm{HF}}_{b\gamma} - \frac{1}{2}\rho^{\mathrm{ee}}_{a\alpha}(\Delta_{\alpha b} - \Delta_{b\alpha})$$
$$= \rho_{\mathrm{eh}}\cdot\Sigma^*_{\mathrm{HF}}|_{ab} - \rho_{\mathrm{ee}}\cdot\Delta|_{ab}$$

and the second one reading

$$-\frac{1}{2}\left(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma}\right)\rho^{\mathrm{eh}}_{\gamma\beta}\rho^{\mathrm{hh}}_{\alpha b} + \left(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma}\right)\rho^{\mathrm{ee}}_{\gamma\alpha}\rho^{\mathrm{eh}}_{\beta b}$$
$$= -\frac{1}{2}\left(\Delta_{\alpha a} - \Delta_{a\alpha}\right)\rho^{\mathrm{hh}}_{\alpha b} + \Sigma^{\mathrm{HF}}_{a\beta}\,\rho^{\mathrm{eh}}_{\beta b}$$
$$= \Delta\cdot\rho_{\mathrm{hh}}|_{ab} + \Sigma_{\mathrm{HF}}\cdot\rho_{\mathrm{eh}}|_{ab}$$

Confirming that the result actually checks out

$$
\begin{aligned}
&\begin{pmatrix} h^{\mathrm{HF}}_{ab} & \Delta_{ab} \\ \Delta^*_{ba} & -h^{\mathrm{HF}}_{ba} \end{pmatrix}\begin{pmatrix} \rho^{\mathrm{ee}}_{ab} & \rho^{\mathrm{eh}}_{ab} \\ \rho^{\mathrm{he}}_{ab} & \rho^{\mathrm{hh}}_{ab} \end{pmatrix} - \begin{pmatrix} \rho^{\mathrm{ee}}_{ab} & \rho^{\mathrm{eh}}_{ab} \\ \rho^{\mathrm{he}}_{ab} & \rho^{\mathrm{hh}}_{ab} \end{pmatrix}\begin{pmatrix} h^{\mathrm{HF}}_{ab} & \Delta_{ab} \\ \Delta^*_{ba} & -h^{\mathrm{HF}}_{ba} \end{pmatrix} \\
&= \begin{bmatrix} \left(h^{\mathrm{HF}}_{ab}\rho^{\mathrm{ee}}_{ab} + \Delta_{ab}\rho^{\mathrm{he}}_{ab}\right) & \left(h^{\mathrm{HF}}_{ab}\rho^{\mathrm{eh}}_{ab} + \Delta_{ab}\rho^{\mathrm{hh}}_{ab}\right) \\ \circ & \circ \end{bmatrix} - \begin{bmatrix} \left(\rho^{\mathrm{ee}}_{ab}h^{\mathrm{HF}}_{ab} - \rho^{\mathrm{eh}}_{ab}\Delta^*_{ba}\right) & \left(\rho^{\mathrm{ee}}_{ab}\Delta_{ab} - \rho^{\mathrm{eh}}_{ab}h^{\mathrm{HF}}_{ba}\right) \\ \circ & \circ \end{bmatrix} \\
&= \begin{bmatrix} \left(\left[h^{\mathrm{HF}}_{ab}, \rho^{\mathrm{ee}}_{ab}\right] + \Delta_{ab}\rho_{\mathrm{he}} + \rho^{\mathrm{eh}}_{ab}\Delta^*_{ba}\right) & \left(h^{\mathrm{HF}}_{ab}\rho^{\mathrm{eh}}_{ab} - \rho^{\mathrm{eh}}_{ab}h^{\mathrm{HF}}_{ba} - \rho^{\mathrm{ee}}_{ab}\Delta_{ab} + \Delta_{ab}\rho^{\mathrm{hh}}_{ab}\right) \\ \circ & \circ \end{bmatrix}
\end{aligned}
$$

$$
\begin{aligned}
&\rho^{\mathrm{hh}}_{ab} = \left\langle c_b c^\dagger_a \right\rangle = -\left\langle c^\dagger_a c_b \right\rangle + \delta_{ab} = -\rho^{\mathrm{ee}}_{ba} + \delta_{ab} \\
&\rho^{\mathrm{he}}_{ab} = \left\langle c^\dagger_b c^\dagger_a \right\rangle = \langle c_a c_b \rangle^\dagger = \left(\rho^{\mathrm{eh}}_{ba}\right)^\dagger \\
&\{c_a, c_b\} = 0 \Leftrightarrow c_a c_b = -c_b c_a \Rightarrow \rho^{\mathrm{eh}}_{ba} = -\rho^{\mathrm{eh}}_{ab} \\
&\left\{c^\dagger_a, c^\dagger_b\right\} = 0 \Leftrightarrow c^\dagger_a c^\dagger_b = -c^\dagger_b c^\dagger_a \Rightarrow \rho^{\mathrm{eh}}_{ba} = -\rho^{\mathrm{eh}}_{ab} \\
&\Delta_{ab} = V^{ab}_{\delta\gamma}\rho^{\mathrm{eh}}_{\gamma\delta} = V^{ba}_{\gamma\delta}\rho^{\mathrm{eh}}_{\gamma\delta} \overset{\delta \rightleftarrows \gamma}{=} V^{ba}_{\delta\gamma}\rho^{\mathrm{eh}}_{\delta\gamma} = V^{ba}_{\delta\gamma}\left(-\rho^{\mathrm{eh}}_{\gamma\delta}\right) = -\Delta_{ba}
\end{aligned}
$$

Putting these two results together yields a Nambu rDM EoM in terms of an effective Bogoliubov-de Gennes (BdG) Hamiltonian as simply as

$$
i\hbar\frac{d}{dt}\rho_{ab} = \left[H^{\mathrm{BdG}}_{ab}, \rho_{ab}\right] \text{ with } H_{\mathrm{BdG}} = \begin{pmatrix} h^{\mathrm{HF}}_{ab} & \Delta_{ab} \\ \Delta^*_{ba} & -h^{\mathrm{HF}}_{ba} \end{pmatrix} \tag{20}
$$

C. Spin-1/2 Hubbard model in the Wannier limit

Furthermore, consider now the approximation where the orbitals are ultra-localized (known as Wannier orbitals) such that the interaction potential, written in terms of an arbitrary electronic basis $\{\phi_\alpha(\mathbf{r})\}$, approximates to

$$
V^{\alpha\beta}_{\gamma\delta} = \int d\mathbf{r}d\mathbf{r}'\phi^*_\alpha(\mathbf{r})\phi^*_\beta(\mathbf{r}')V(\mathbf{r}-\mathbf{r}')\phi_\gamma(\mathbf{r}')\phi_\delta(\mathbf{r}) \approx V^{\alpha\beta}\delta_{\alpha\delta}\delta_{\beta\gamma} \tag{21}
$$

with $V^{\alpha\beta} \equiv V(\mathbf{r}_\alpha - \mathbf{r}_\beta)$. Within the scope of this approximation the interaction Hamiltonian will becomes simply $H_{\mathrm{int}} = \frac{1}{2}\sum_{\alpha\beta} V^{\alpha\beta}c^\dagger_\alpha c^\dagger_\beta c_\beta c_\alpha$. The Hartree and Fock self-energies become

$$
\begin{aligned}
\Sigma^H_{ab} &= \textstyle\sum_\gamma V^{a\gamma}\rho^{\mathrm{ee}}_{\gamma\gamma}\delta_{ab} \\
\Sigma^F_{ab} &= -V^{ab}\rho^{\mathrm{ee}}_{ab}
\end{aligned}
\quad \text{and} \quad \Delta_{ab} = V^{ab}\rho^{\mathrm{eh}}_{ab} \tag{22}
$$

Furthermore, for the sake of getting the problem in a tight-binding scheme, we isolate the degrees of freedom into purely spatial and spin indices $a \to i\sigma_i$ such that

$$
\boldsymbol{\rho}_{ij} = \begin{pmatrix} \boldsymbol{\rho}^{\mathrm{ee}}_{ij} & \boldsymbol{\rho}^{\mathrm{eh}}_{ij} \\ \boldsymbol{\rho}^{\mathrm{he}}_{ji} & \boldsymbol{\rho}^{\mathrm{hh}}_{ji} \end{pmatrix} = \left(\begin{array}{cc|cc} \rho^{\mathrm{ee}}_{i\uparrow j\uparrow} & \rho^{\mathrm{ee}}_{i\uparrow j\downarrow} & \rho^{\mathrm{eh}}_{i\uparrow j\uparrow} & \rho^{\mathrm{eh}}_{i\uparrow j\downarrow} \\ \rho^{\mathrm{ee}}_{i\downarrow j\uparrow} & \rho^{\mathrm{ee}}_{i\downarrow j\downarrow} & \rho^{\mathrm{eh}}_{s\downarrow\uparrow} & \rho^{\mathrm{eh}}_{i\downarrow j\downarrow} \\ \hline \rho^{\mathrm{he}}_{i\uparrow j\uparrow} & \rho^{\mathrm{he}}_{i\uparrow j\downarrow} & \rho^{\mathrm{hh}}_{i\uparrow j\uparrow} & \rho^{\mathrm{hh}}_{i\downarrow j\uparrow} \\ \rho^{\mathrm{he}}_{i\downarrow j\uparrow} & \rho^{\mathrm{he}}_{i\downarrow j\downarrow} & \rho^{\mathrm{hh}}_{i\uparrow j\downarrow} & \rho^{\mathrm{hh}}_{i\downarrow j\downarrow} \end{array}\right) \tag{23}
$$

Note the use of the bold to denote that an object possesses a spin structure. Moreover, notice that, since we considered the interaction to be spin-independent, we also have that $V^{i\sigma_i j\sigma_j} \equiv V^{ij}$. In this basis, the self-energies read as

$$
\begin{aligned}
\Sigma^H_{i\sigma_i j\sigma_j} &= \sum_s V^{is}\left(\rho^{\mathrm{ee}}_{s\uparrow s\uparrow} + \rho^{\mathrm{ee}}_{s\downarrow s\downarrow}\right)\delta_{ij}\delta_{\sigma_i\sigma_j} \\
\Sigma^F_{i\sigma_i j\sigma_j} &= -V^{ij}\rho^{\mathrm{ee}}_{i\sigma_i j\sigma_j} \text{ and } \Delta_{i\sigma_i j\sigma_j} = V^{ij}\rho^{\mathrm{eh}}_{i\sigma_i j\sigma_j}
\end{aligned} \tag{24}
$$

Notice that the Hartree term is exclusively an onsite due to the δ_{ij}. If we now define a new object related to the Nambu rDM but specifically having the same symmetries as the Nambu Hamiltonian, this being

$$\tilde{\boldsymbol{\rho}}_{ij} = \begin{pmatrix} \boldsymbol{\rho}_{ij}^{\mathrm{ee}} & -\boldsymbol{\rho}_{ij}^{\mathrm{eh}} \\ -\left(\boldsymbol{\rho}_{ji}^{\mathrm{eh}}\right)^{*} & -\boldsymbol{\rho}_{ji}^{\mathrm{ee}} \end{pmatrix}, \tag{25}$$

we can then write the total self-energy of the system in an elegant and compact form as

$$\boldsymbol{\Sigma}_{ij} = \frac{1}{2}\sum_{s} V^{is}\mathrm{Tr}\left(\left[\tau_z \otimes \sigma_0\right]\tilde{\boldsymbol{\rho}}_{ss}\right)\left[\tau_z \otimes \sigma_0\right]\delta_{ij} - V^{ij}\tilde{\boldsymbol{\rho}}_{ij}, \tag{26}$$

with both σ and τ as Pauli matrices but operating in their respectively spin and electron-hole subspaces the Hamiltonian. In the limiting case of a spin-1/2 Hubbard model one considers $V^{ij} = U\delta_{ij}$ such that mean field decoupling gives rise to just the onsite term,

$$\boldsymbol{\Sigma}_{ij} = \frac{1}{2}U\left[\mathrm{Tr}\left(\left[\tau_z \otimes \sigma_0\right]\tilde{\boldsymbol{\rho}}_{ii}\right)\left[\tau_z \otimes \sigma_0\right] - 2\tilde{\boldsymbol{\rho}}_{ij}\right]\delta_{ij}.$$

Hubbard self-energy

$$\begin{aligned}
\boldsymbol{\Sigma}_i =& \begin{pmatrix} \sum_s V^{is}\left(\rho_{s\uparrow\uparrow}^{\mathrm{ee}} + \rho_{s\downarrow\downarrow}^{\mathrm{ee}}\right)\mathbb{1} - V^i\boldsymbol{\rho}_i^{\mathrm{ee}} & V^i\boldsymbol{\rho}_i^{\mathrm{eh}} \\ \left[V^i\boldsymbol{\rho}_i^{\mathrm{eh}}\right]^T & -\sigma_y\left[\sum_s V^{is}\left(\rho_{s\uparrow s\uparrow}^{\mathrm{ee}} + \rho_{s\downarrow s\downarrow}^{\mathrm{ee}}\right)\mathbb{1} - V^i\boldsymbol{\rho}_i^{\mathrm{ee}}\right]^T\sigma_y \end{pmatrix} \\
=& \frac{1}{2}\sum_s V^{is}\mathrm{Tr}\left(\tau_z\tilde{\boldsymbol{\rho}}_s\right)\begin{pmatrix} \mathbb{1} & \\ & -\mathbb{1} \end{pmatrix} - V^i\begin{pmatrix} \boldsymbol{\rho}_i^{\mathrm{ee}} & -\boldsymbol{\rho}_i^{\mathrm{eh}} \\ -\boldsymbol{\rho}_i^{\mathrm{eh}} & -\sigma_y\boldsymbol{\rho}_i^{\mathrm{ee}}\sigma_y \end{pmatrix} \\
=& \frac{1}{2}\sum_s V^{is}\mathrm{Tr}\left(\tau_z\tilde{\boldsymbol{\rho}}_s\right)\left[\tau_z \otimes \sigma_0\right] - V^i\tilde{\boldsymbol{\rho}}_i
\end{aligned}$$

where the trace reads as

$$\mathrm{Tr}\left(\tau_z\bar{\boldsymbol{\rho}}_s\right) = \mathrm{Tr}\left(\begin{array}{cc|cc} \rho_{s\uparrow\uparrow}^{\mathrm{ee}} & \rho_{s\uparrow\downarrow}^{\mathrm{ee}} & -\rho_{s\uparrow\uparrow}^{\mathrm{eh}} & -\rho_{s\uparrow\downarrow}^{\mathrm{eh}} \\ \rho_{s\downarrow\uparrow}^{\mathrm{ee}} & \rho_{s\downarrow\downarrow}^{\mathrm{ee}} & -\rho_{s\downarrow\uparrow}^{\mathrm{eh}} & -\rho_{s\downarrow\downarrow}^{\mathrm{eh}} \\ \hline -\left[\rho_{s\uparrow\uparrow}^{\mathrm{he}}\right] & -\left[\rho_{s\uparrow\downarrow}^{\mathrm{he}}\right] & -\left[-\rho_{s\uparrow\uparrow}^{\mathrm{ee}}\right] & -\left[\rho_{s\downarrow\uparrow}^{\mathrm{ee}}\right] \\ -\left[\rho_{s\downarrow\uparrow}^{\mathrm{he}}\right] & -\left[\rho_{s\downarrow\downarrow}^{\mathrm{he}}\right] & -\left[\rho_{s\uparrow\downarrow}^{\mathrm{ee}}\right] & -\left[-\rho_{s\downarrow\downarrow}^{\mathrm{ee}}\right] \end{array}\right) = 2\left(\rho_{s\uparrow\uparrow}^{\mathrm{ee}} + \rho_{s\downarrow\downarrow}^{\mathrm{ee}}\right)$$